Equivalent & Simplified Ratios

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Equivalent ratios

What is an equivalent ratio?

- **Equivalent ratios** are two ratios that represent the **same proportion** of quantities within a whole
  - E.g. The ratio **5 : 10** is equivalent to **20 : 40**
- Equivalent ratios are frequently used when the values involved take on a **real-life meaning**
  - E.g. A cake recipe involves flour and butter being mixed in the ratio 3 : 2
    - 3 g of flour and 2 g of butter would not lead to a very big cake
    - An equivalent ratio of 300 : 200 gives a more realistic 300 g of flour and 200 g of butter

How do I find an equivalent ratio?

- You can find an **equivalent ratio** by **multiplying** (or **dividing**) **each part** of the ratio by the **same value**
  - E.g. Multiply each part of the ratio 2 : 3 : 7 by 4 to find an equivalent ratio of 8 : 12 : 28
  - Ratios can be scaled up or down to suit the context of a question
- The size of each part in the ratio, **relative** to the others, is still the same
  - The **actual values** in the equivalent ratio may be **more meaningful** in the context of the situation
- Finding an equivalent ratio is **similar** to finding **equivalent fractions**
  - However it is crucial to remember that 1 : 4 is **not** equivalent to $\frac{1}{4}$
- In questions you will be given **information** that will allow you to find the **multiplier**
  - The information may be about the one part of the ratio, or it may be about the whole
  - You can then use this to find the other parts of the equivalent ratio and answer the question

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Help me understand

Writing down what you are **doing** to each part of the ratio helps show your working and makes it easier to keep track of what you are doing.

E.g.

$$\begin{array}{ccc} & A & : & B \\ & 3 & : & 4 \\ \times 5 \downarrow & & & \downarrow \times 5 \\ & 15 & : & 20 \end{array}$$

Worked Example

The ratio of cabbage leaves eaten by two rabbits, Alfred and Bob, is 7 : 5.

(a) Write down an equivalent ratio that would involve a total of 48 cabbage leaves being eaten.

Answer:

We have information about the whole so first add up the parts of the ratio to find how many parts make the whole

$$7 + 5 = 12$$

We require a total of 48 so divide this by 12 to find the multiplier

$$48 \div 12 = 4$$

Answer the question by multiplying each part of the ratio by this multiplier

$$\begin{array}{ccc} & A & : & B \\ & 7 & : & 5 \\ \times 4 \downarrow & & & \downarrow \times 4 \\ & 28 & : & 20 \end{array}$$

28 : 20

(b) On another occasion Bob eats 35 leaves. Find out how many leaves Alfred eats.

Answer:

Use the information about Bob to find the multiplier using division

$$35 \div 5 = 7$$

Now multiply Alfred's part of the ratio by the multiplier to answer the question

$$\begin{array}{ccc} & A & : & B \\ & 7 & : & 5 \\ \times 7 \downarrow & & & \downarrow \times 7 \\ & 49 & : & 35 \end{array}$$

Simplifying ratios

What is a simplified ratio?

- **Simplifying a ratio** involves finding an **equivalent ratio** where the numbers involved are **smaller**
  - E.g. The ratio **45 : 30** is equivalent to **9 : 6**
- The need to **simplify** a ratio often arises when the initial ratio has been given using **large values** from a **real-life context**
- A ratio is in its **simplest form** when
  - All of the values in the ratio are **integers**
  - There are no **common factors** between each of the values in the ratio
  - E.g. The simplest form of the ratio **45 : 30** is **3 : 2**

How do I simplify a ratio?

- **Divide each part** of the ratio by the **same value**
  - This value should be a **common factor** of all parts of the ratio
    - Ideally, the **highest common factor** **(HCF)** should be used to get the ratio into its simplest form in one go
    - If the HCF is not used, we can repeat the process of simplifying
  - E.g. Divide all parts of the ratio **30 : 66 : 12** by 6 to find the ratio in its simplest form **5 : 11 : 2**

Examiner Tips and Tricks

Make the most of your **calculator** in the exam.

If you type the ratio in as though it was a fraction (although remember it is **not**) your calculator will give you the answer in its simplified form.

Worked Example

Amber and Naomi are sharing a large cake that has been cut into 48 pieces.
Amber receives 30 of these pieces, Naomi receives the rest.

Write down the ratio, in its simplest form, of the number of pieces of cake that Amber receives to the number of pieces of cake that Naomi receives.

Answer:

$48 - 30 = 18$

Next write down the ratio without simplifying
Make sure you have the order correct ('Amber to Naomi')

$$30 : 18$$

Look for a common factor of both 30 and 18 to simplify the fraction
6 is the highest common factor
Divide both parts of the fraction by 6

$$\begin{array}{ccc} A & : & N \\ 30 & : & 18 \\ \div 6 \downarrow & & \downarrow \div 6 \\ 5 & : & 3 \end{array}$$

5 : 3

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